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Mechanical Engineering, Robotics Minor · Texas Tech University · Expected May 2028

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What I enjoy most in engineering is the loop between design and testing: turning an idea into a CAD model, proving it with simulation, then building the real thing and pushing it until I learn where it holds up and where it doesn't. Every project here follows that loop, from a 3D-printed quadcopter frame I designed and flew to autonomous drones in the VILAI Lab.

MOST INTERESTED IN **Design · Testing · Troubleshooting · Hands-on hardware**

PROJECTS

- 01

Custom Quadcopter Frame

Designed the frame to a 2.5 safety-factor target (4.2 in FEA), built the electronics, and flew it.

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- 02

Portable Door Lock

Team project that met every written requirement under a \$20 budget, then fitted to a real door.

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- 03

Three-Car Train & Track Assembly

Designed and toleranced every part, then proved the mechanism with a motion study.

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- 04

Autonomous Quadrotor Integration: ModalAI Starling 2 Max

VILAI Lab research: bringing up a PX4 / VOXL 2 autonomous drone from bench setup to flight testing.

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TOOLS & METHODS

CAD & ANALYSIS	SolidWorks	Fusion 360	FEA / stress simulation	Motion studies	Tolerancing
HARDWARE	3D printing	Soldering	Arduino / Teensy	UAV integration	
FLIGHT SYSTEMS	PX4	ModalAI VOXL	dRehmFlight		
SOFTWARE	Python				

01 Quadcopter Frame — Design & Stress Test

Individual project · July 2026 – present · Fusion 360 · FEA · 3D printing

Goal: design my own compact quadcopter frame that carries the flight electronics and battery, prove in simulation that the arms can take flight loads with a safety factor of at least 2.5, and only then print and fly it.

01 – DESIGN

Modeled in Fusion 360

2 frame revisions

02 – TEST

Static stress study

Target SF 2.5 → result 4.2

03 – BUILD

Printed, wired, flown

Detailed on page 3



FIG 1 My frame geometry set up for analysis: four arms off a stacked center body, supports at the motor mounts and the payload load at the center.

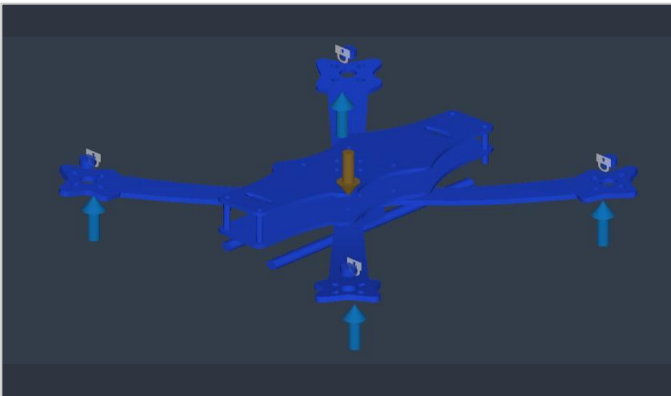
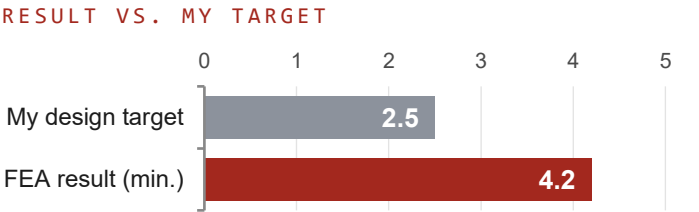


FIG 2 Safety factor result across the frame. No region fell below my 2.5 target, so the design went to printing without adding material.

- WHAT I DESIGNED
- Drew every part of the frame myself in Fusion 360 — arms, stacked center plates, standoffs and motor mounts — then joined them into one assembly.
 - Sized the center stack around the flight controller, ESC and a 4S battery.
 - Kept the arms as short as the propellers allowed so the airframe stayed small and light.
 - Revised the frame twice, using simulation results to decide where material could come out.

LOAD CASE

INPUT	VALUE
Thrust	10 N up at each motor mount (40 N total)
Weight	9.81 N down at the center body (≈ 1 kg)
Material	ABS (simulated)



Target is the requirement I set before modeling; the result is the lowest safety factor anywhere on the frame.

WHAT THE ANALYSIS TOLD ME

The frame cleared my 2.5 target with margin, so the structure was not the limiting part of the design — the electronics were. That is what page 3 focuses on. Two caveats I keep in mind: the study treats the frame as solid material, while a 3D print is weaker along its layer lines; and I simulated ABS but printed in PLA to stay on budget. PLA holds up about as well under steady loads but is more brittle, so the real margin in a hard landing or crash is lower than the number above. A fairer check would re-run the study with PLA properties and an impact load.

01 Quadcopter — Electronics & Build

Component selection · Hand soldering · Firmware setup · Ground and flight testing

Once the frame passed analysis, I selected the electronics, laid out how power and control signals move through the aircraft, and wired it by hand. This is the system I built and flew.

SYSTEM ARCHITECTURE I BUILT

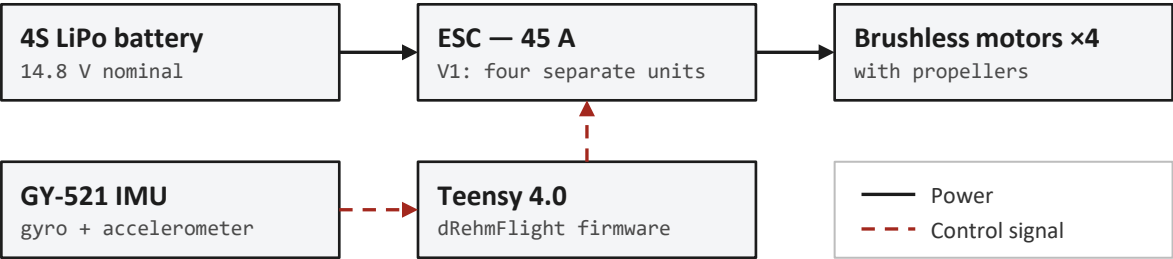


FIG 3 The battery feeds the ESCs, which drive the motors. The IMU reports attitude to the Teensy, which corrects each motor's throttle in a closed loop.

COMPONENT	ROLE IN THE SYSTEM
4S LiPo battery	Single power source, sized for the current all four motors draw at full throttle.
45 A ESCs	Convert battery DC into the three-phase drive a brushless motor needs; set each motor's speed.
Brushless motors ×4	Produce thrust. Speed differences between them tilt and turn the aircraft.
GY-521 IMU	Measures rotation rate and acceleration so the controller knows how the airframe is oriented.
Teensy 4.0	Runs dRehmFlight: reads the IMU and adjusts each motor's throttle to hold the aircraft stable.



FIG 4 The build on the bench: four motors wired back to the ESCs, with the capacitor and battery lead soldered in place.

HOW I BUILT IT

- Soldered every power and signal connection by hand.
- Fitted the stack into the center body I had designed for it.

HOW I TESTED IT

- Confirmed ESC startup tones before arming anything.
- Checked each motor's response and spin direction on the bench.
- Ran incremental ground tests, then flew it.

HOW I SET UP THE FIRMWARE (DREHMFLIGHT)

IMU setup Teensy reads the GY-521 gyro and accelerometer to estimate attitude.	Motor mixing Quad-X mixing, with each motor on its own output channel.	Spin direction Alternating rotation so opposing pairs cancel propeller torque.	Pilot input Throttle, roll, pitch and yaw mapped to attitude commands.
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RESULT — V1

The aircraft I designed, printed and wired got off the ground and flew. A later crash damaged the separate ESCs, which set up the next revision.

IN PROGRESS — V2

Rebuilding around a single 4-in-1 ESC: it suits a frame this small, costs less than replacing four units, and cuts the number of hand-soldered joints. Goal is fully controllable flight.

02 Portable Door Lock

Team design project, Texas Tech University · November 2025 · SolidWorks · 3D printing

Goal: a lock you can carry in a bag and fit to a public stall door that does not lock. The threaded bolt passes through the gap at the door edge, a disc clamps each side, and the outward face reads IN USE.

REQUIREMENTS THE TEAM DESIGNED TO

REQUIREMENT	TARGET	RESULT
Load capacity	Hold against 22.2 N (5.0 lb) without slipping	✓ Met
Clearance	Threaded bolt ≤ 8.4 mm so it fits most stall door gaps	✓ Met
Durability	At least 50 locking cycles with no loss of performance	✓ Met
Portability	Envelope of 88.9 × 40.6 mm so it stays compact	✓ Met
Budget	Total project cost under \$20	✓ Met

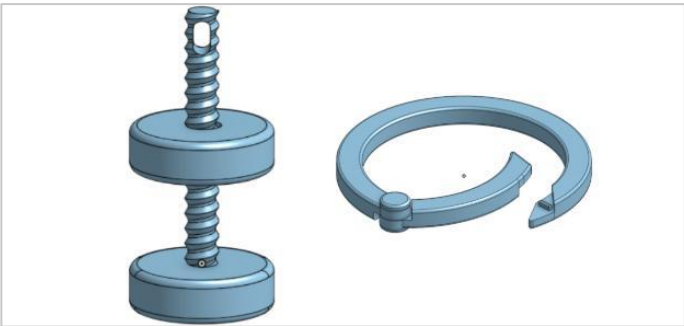


FIG 1 SolidWorks model: the threaded bolt with its two clamping discs, and the split ring that tightens and carries it.



FIG 2 The printed parts: the IN USE cap, the second disc with its threaded bore, and the split ring.



FIG 3 Fitted to a stall door: the bolt sits in the gap with a disc clamped each side, and the ring gives grip to tighten it by hand.



FIG 4 The same lock from outside the stall — the face that shows carries the status the user wants read.

MY ROLE

Helped design the parts in SolidWorks, printed the prototypes, and tracked spending to keep the team under its \$20 budget.

RESULT

A working prototype, printed and assembled from three parts, that met every requirement above when fitted to a real stall door. The ring doubles as the grip and the carry loop.

PROCESS

Worked as a team from written requirements, through SolidWorks modeling, to two printed prototypes before this final design — then presented it in a design review with peers and faculty.

03 Three-Car Train & Track Assembly

Engineering Graphics, Texas Tech University · Spring 2026 · SolidWorks · Motion study

Goal: design a train that works as a real mechanism — every part modeled from scratch, three cars coupled into one assembly on a track, with wheels that turn and drive the train forward.

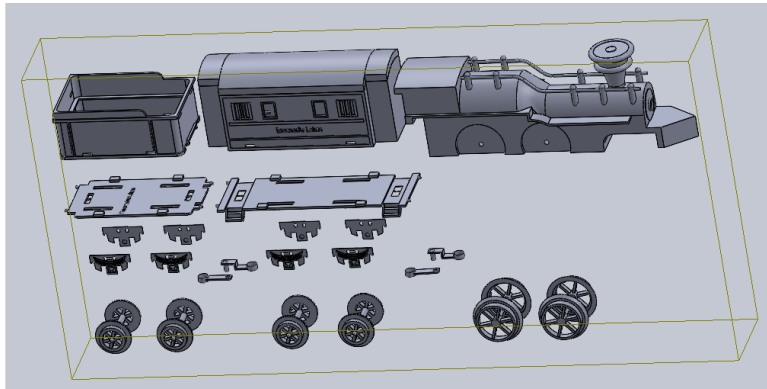


FIG 1 Exploded view of the parts I modeled: three car bodies, chassis plates, bogie frames, couplers and wheelsets.

01 — WHAT I DESIGNED

- Modeled every component as its own part file — bodies, chassis plates, bogies, couplers, axles and wheels — then built them into car sub-assemblies.
- Designed the detail features on each body, including the locomotive's boiler, stack and running gear.
- Built the track as its own part so the wheels had a real surface to run on.
- Set ± 0.01 mm tolerances on mating features so wheels turn on their axles and couplers connect without binding.

02 — ASSEMBLY STRUCTURE

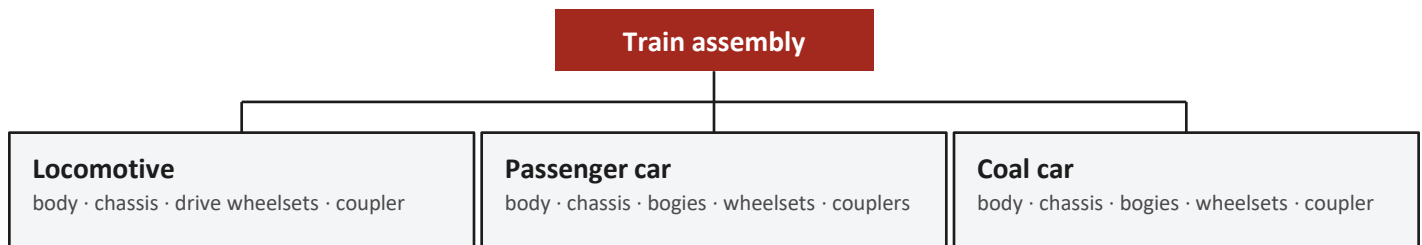


FIG 2 How the assembly is organized: each car is its own sub-assembly, joined to the next by a coupler and mated to the track.

03 — MOTION TEST

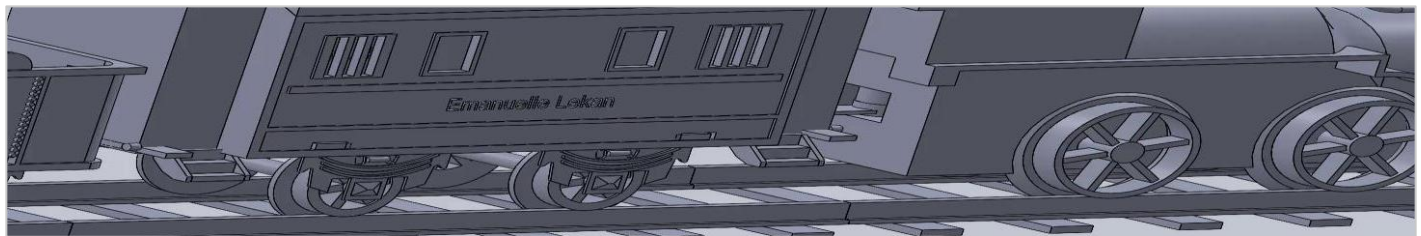


FIG 3 Motion study: rotating the wheels drives all three coupled cars along the track.

► Watch the motion study: [HERE](#)

RESULT

The motion study ran without interference: the wheels turned on their axles, the couplers held, and the whole train tracked along the rails. That confirmed the parts and tolerances I chose actually worked together.

WHAT I WOULD CHANGE

± 0.01 mm is tighter than a printed or machined part would hold. If this were made rather than modeled, I would open the wheel-to-axle clearance to something a real process can hit.

04 Autonomous Quadrotor Integration: Starling 2 Max

VILAI Lab, Texas Tech University · Fall 2026 · ModalAI VOXL 2 · PX4 · VIO

Goal: bring up a ModalAI Starling 2 Max, a research drone built for autonomous, GPS-denied flight, through the ModalAI VOXL Developer Bootcamp, and take it from bench setup to flight testing as part of the lab team.



FIG 1 The Starling 2 Max with its Commando 8 transmitter. Visible: the GNSS antenna on its mast, the VOXL 2 flight computer at the center, and the camera module.

WHAT I WORKED ON

- Working through the ModalAI VOXL Developer Bootcamp to bring the aircraft from bench setup to flight readiness.
- Configuring the VOXL 2 flight computer and the PX4 autopilot that runs on it.
- Completed hardware bring-up and pre-flight checks, including binding the Commando 8 RC transmitter.
- Troubleshot software, firmware and hardware integration issues across the flight stack; traced the last blocker to a licensed VOXL software component and coordinated access through the lab PI.
- Working alongside 11 other undergraduate researchers across majors on testing, equipment and flight readiness.

HOW THE AUTONOMY STACK FITS TOGETHER

01 · SENSE

Cameras, IMU, GNSS

Onboard cameras and IMU observe motion; GNSS adds global position outdoors.

02 · ESTIMATE

Visual-inertial odometry

Fuses camera and IMU data to estimate position without relying on GPS.

03 · CONTROL

PX4 autopilot

Position and attitude loops turn the state estimate into motor commands.

04 · ACTUATE

ESC + 4 motors

Motor speed differences hold, tilt and turn the aircraft.

PLATFORM

COMPONENT	ROLE IN THE SYSTEM
ModalAI VOXL 2	Flight computer that runs PX4 alongside ModalAI's autonomy and vision software on one board.
PX4 autopilot	Flight control: estimation, attitude and position control loops, flight modes and failsafes.
Camera module + IMU	Inputs for visual-inertial odometry, which lets the drone hold position in GPS-denied spaces.
GNSS receiver	Global position fix for outdoor flight.
Commando 8 transmitter	Manual pilot input and flight-mode switching; the safety link during testing.

STATUS

Hardware bring-up and pre-flight checks are complete. The aircraft is being prepared for its first flight tests with the lab team.

WHAT THIS ADDS

Where my own quadcopter (project 01) runs a hand-wired Teensy controller, this platform is an integrated autonomy stack. The work is systems integration: getting sensors, estimation, control and software to run together.